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Predicting Protein-Protein Interaction (PPI) Interfaces from AlphaFold-Predicted Structures Using Graph Neural Networks (GNN)

Summary

This project investigates whether GNN models trained to predict PPI interfaces from experimental crystal structures maintain their accuracy when applied to AlphaFold-predicted structures. I propose a systematic benchmarking of the PeSTo geometric transformer on matched experimental and AlphaFold structures, followed by an error analysis correlating prediction failures with AlphaFold's per-residue confidence scores (pLDDT). The goal is to quantify the performance gap and identify the structural features responsible, laying the groundwork for confidence-aware interface predictors suitable for proteome-scale analysis.

Motivation and Background

Protein-protein interactions are central to virtually all cellular functions. From signal transduction cascades to immune recognition, the specificity of biological processes depends on which protein surfaces come into contact and how tightly they bind.

Disruptions to these interactions, whether through mutations at interface residues, conformational changes, or the loss of binding partners, can derail cellular pathways and contribute to diseases including cancer and neurodegeneration [1]. Understanding which residues participate in binding interfaces is therefore fundamental to both basic biology and drug design.

Experimental approaches for characterizing PPI interfaces, including X-ray co-crystallography, cryo-electron microscopy of complexes, and cross-linking mass spectrometry, provide high-resolution structural detail but are expensive, time-consuming, and limited in throughput. The Protein Data Bank (PDB) contains roughly 220,000 experimentally determined structures, but this represents only a small fraction of the proteins encoded across known genomes. The arrival of AlphaFold changed the landscape dramatically: the AlphaFold Protein Structure Database now houses over 200 million predicted structures [2], providing structural coverage for essentially all known protein sequences. If computational interface predictors could operate reliably on these predicted structures, it would enable interaction mapping at an unprecedented scale.

Several deep learning methods have recently achieved impressive accuracy for predicting PPI binding interfaces from protein structure. PeSTo (Protein Structure Transformer) is a geometric transformer that acts directly on atomic coordinates labeled only with element names, requiring no handcrafted physicochemical features [3]. Trained on all biological

assemblies from the PDB, PeSTo can predict not only protein-protein interfaces but also binding sites for nucleic acids, lipids, ions, and small molecules. ScanNet takes a complementary approach, using an interpretable geometric deep learning architecture to identify interface residues and antibody epitopes from solved structures [4]. HIGH-PPI introduced a hierarchical graph framework that models both residue-level interactions within proteins and protein-level interactions between them, capturing the natural multi-scale organization of PPI networks [5]. All these methods represent proteins as graphs—nodes for residues or atoms, edges for spatial proximity—and learn to classify surface residues as belonging to an interface or not.

A critical but under-examined assumption in this field is that models trained on high-resolution experimental structures will generalize to predicted structures. AlphaFold predictions, while highly accurate on average, are not uniformly reliable. AlphaFold's own per-residue confidence metric, pLDDT, classifies residues with scores above 90 as very high confidence, 70-90 as confident, 50-70 as low confidence, and below 50 as very low confidence (often corresponding to intrinsic disorder) [2, 6]. In the human proteome, only 36% of residues are modeled with very high confidence (pLDDT > 90), and the regions with the lowest confidence tend to be surface-exposed loops and flexible termini [7]—precisely the regions where many protein-protein binding events occur. AlphaFold has also been shown to sometimes over-predict secondary structure in loop regions [8], producing geometries that differ from native conformations in ways that could mislead interface predictors.

Recent work by Yuan et al. directly quantified this concern [9]. In their development of GPSite, a protein language model-based binding site predictor, they benchmarked PeSTo and ScanNet on both experimental structures and structures predicted by ESMFold (a language model-based structure predictor comparable to AlphaFold). PeSTo's area under the precision-recall curve (AUPR) dropped from 0.797 on experimental structures to 0.691 on predicted structures, and ScanNet's AUPR fell from 0.476 to 0.399 [9]. These are substantial declines that call into question the reliability of applying current interface predictors to the AlphaFold proteome without further validation or adaptation.

Central Question

The central question of this project is: how much accuracy do structure-based GNN interface predictors lose when applied to AlphaFold-predicted structures instead of experimental ones, and can we identify the structural features that explain this degradation? We address this through two complementary analyses: a controlled benchmarking study and a feature-level error analysis. We then outline a proposed adaptation strategy informed by the results.

Benchmarking on matched experimental and predicted structures. To isolate the effect of structural quality on interface prediction, we would construct a carefully matched dataset. Starting from proteins in the PDB that participate in characterized complexes with interface annotations available through the PISA (Proteins, Interfaces, Surfaces, and Assemblies) service [10], we would select cases where both a high-resolution experimental structure (≤ 2.5 Å resolution) and a corresponding AlphaFold-predicted structure are available. This paired design is critical: by running PeSTo on the experimental and predicted structure of the same protein and comparing predictions against the same ground-truth interface labels, any difference in performance can be attributed to the structural input rather than differences in protein difficulty.

Performance would be evaluated using two standard metrics: ROC-AUC, which measures discrimination between interface and non-interface residues across all classification thresholds, and AUPR, which is more informative when the classes are imbalanced (as they are here, since most surface residues are not part of an interface). Based on the findings of Yuan et al. [9], we anticipate a measurable drop in AUPR on AlphaFold structures. Their study used ESMFold-predicted structures; since AlphaFold generally produces more accurate predictions than ESMFold [11], the degradation on AlphaFold structures might be somewhat smaller, but the direction of the effect should be consistent. The magnitude of this gap is the key quantity we aim to establish.

Error analysis: linking prediction failures to structural confidence. The second component is a systematic analysis of where and why the predictions fail on AlphaFold structures. For every residue in the test set, we would record: (a) whether the interface prediction is correct or incorrect, (b) the AlphaFold pLDDT score at that position, and (c) the local RMSD between the predicted and experimental structure in a neighborhood around that residue.

We hypothesize that prediction errors will be enriched in regions with low pLDDT scores (below 70). This hypothesis is grounded in two observations from the literature. First, AlphaFold's official documentation states that regions with pLDDT below 70 have low confidence, with backbone positions potentially deviating from the true structure [6]. Second, a Nature Methods analysis found that even high-confidence AlphaFold predictions can show inter-atomic distance deviations of 0.5–1 Å at longer range compared to experimental structures, with moderate-to-high-confidence regions (pLDDT > 70) exhibiting deviations of roughly 0.1 Å for nearby atom pairs and 0.7 Å for distant pairs [7]. Since interface prediction relies heavily on the precise geometry of surface residues, even modest coordinate shifts in loop regions could flip a borderline classification.

To test this, we would stratify the residues by pLDDT confidence band and compute separate ROC-AUC and AUPR values for each stratum. We would also compute the

Pearson correlation between local RMSD (experimental vs. AlphaFold) and the per-residue prediction error magnitude. If the correlation is strong and negative (higher RMSD correlates with worse predictions), this would directly implicate structural inaccuracy as the primary driver of the performance gap, rather than other factors such as systematic biases in AlphaFold’s representation of surface electrostatics or hydrophobicity.

Proposed adaptation: pLDDT-augmented graph features. If the benchmarking and error analysis confirm a meaningful, pLDDT-correlated performance gap, we would propose a straightforward adaptation strategy. PeSto’s architecture operates on atomic coordinates labeled with element types. We would augment the node features in the residue graph with the AlphaFold pLDDT score as an additional input channel, then fine-tune the pre-trained model on a mixed dataset containing both experimental structures (with pLDDT set to 100, representing perfect confidence) and AlphaFold-predicted structures (with their actual pLDDT values). This approach requires no architectural changes to PeSto—it merely gives the model access to information about structural reliability that it currently lacks.

The rationale is that a confidence-aware model could learn to down-weight interface predictions in regions where the input geometry is unreliable, rather than treating all atomic coordinates as equally trustworthy. An alternative approach—trimming low-confidence regions before prediction, as proposed by Tam et al. for general AlphaFold applications [12]—risks discarding precisely the residues we most need to classify, since many binding events involve flexible loops. By incorporating confidence as a feature rather than a filter, the model retains the ability to make predictions everywhere while appropriately calibrating its certainty.

Context from the broader PPI prediction landscape. It is worth noting that this question—transferability from experimental to predicted structures—is not unique to PPI interface prediction. In drug discovery, a study on G-protein-coupled receptors found that while AlphaFold models captured binding pocket geometry more accurately than traditional homology models, docking predictions using AlphaFold structures were substantially less accurate than those using experimental co-crystal structures [13]. Similarly, an evaluation of AlphaFold3-predicted complexes for binding free energy prediction found an 8.6% increase in RMSE compared to using experimental PDB structures [14]. These findings from adjacent fields reinforce the premise that structural prediction accuracy matters for downstream computational analyses, and that the gap needs to be explicitly measured and addressed for each application domain.

Outlook

This work would establish the first systematic, quantitative baseline for how well geometric deep learning models for PPI interface prediction transfer from experimental structures to the AlphaFold structural proteome. The immediate practical outcome would be a set of guidelines for when AlphaFold structures can be trusted as inputs for interface prediction (e.g., above a certain pLDDT threshold) and when caution is warranted.

Looking further ahead, the pLDDT-augmented fine-tuning approach, if successful, could serve as a general-purpose preprocessing strategy applicable to any geometry-based interface predictor, not just PeSTo. The same principle of incorporating structural confidence into the prediction pipeline could extend to other tasks that depend on surface geometry, such as epitope prediction for vaccine design, molecular docking, and protein-nucleic acid interaction prediction. The dream experiment would be to apply a validated, confidence-aware interface predictor across the entire human AlphaFold proteome to generate a draft “interfaceome”, or a comprehensive map of all predicted binding surfaces across all human proteins and compare it against known interaction networks to discover novel, previously uncharacterized binding interfaces. PeSTo’s authors have already begun exploring this direction with high-quality AlphaFold models [3], but a confidence-aware version could extend this analysis to the many proteins whose AlphaFold models contain substantial low-confidence regions, greatly expanding coverage.

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